

Graph Enhanced Financial RAG System

Corporate financial reports, particularly SEC 10-K filings, contain extensive textual and numerical information spread across multiple sections, making accurate financial question answering a challenging retrieval and reasoning task. Conventional Retrieval-Augmented Generation (RAG) systems primarily depend on vector-based similarity search over unstructured passages, which often proves insufficient for complex queries requiring reasoning across multiple entities, fiscal years, and interconnected financial concepts. Graph-based RAG overcomes this by organizing information as structured knowledge, while Agentic RAG further enhances retrieval by enabling a language model to plan, select, and coordinate retrieval strategies according to query needs. This project proposes an agentic GraphRAG system for financial question answering built on the FinReflectKG knowledge graph, that dynamically routes each query through lightweight or comprehensive multi-tool retrieval based on estimated reasoning complexity, rather than applying a fixed retrieval strategy uniformly. The system integrates graph traversal, hybrid document retrieval, and symbolic financial computation to retrieve evidence, perform calculations such as growth rates and ratios, and generate traceable responses. Preliminary evaluation across staged configurations — no-retrieval, vector RAG, graph-enriched RAG, and agentic GraphRAG — on the FinReflectKG-MultiHop benchmark reveals a gap between retrieval quality and reasoning accuracy in graph-enriched settings, which this work addresses through structured evidence formatting and adaptive routing. The framework targets improved temporal and cross-company reasoning while reducing computational overhead relative to uniformly comprehensive retrieval. Evaluation metrics include answer accuracy, retrieval efficiency, latency, and evidence traceability, with the goal of a financially accurate, explainable, and computationally efficient question-answering system.